Point Label Meets Remote Sensing Change Detection: A Consistency-Aligned Regional Growth Network

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***Abstract*— The acquisition of a substantial volume of precisely dense pixel-annotated samples plays a crucial role in the effective training of deep learning-based change detection models. Nevertheless, in real-world scenarios, pairwise labeling of massive bitemporal remote sensing images is often laborious and time-consuming, resulting in the lack of labeled samples. In this article, we propose a novel point-based weakly supervised learning approach, called as the consistency-aligned regional growth network (CARGNet), for remote sensing change detection. Unlike pixel-level labels, point labels are easy to label and usually sparse, which leads to a lack of boundary information, making it difficult for the model to accurately capture the details of the changed objects. Therefore, learning directly from them may mislead the training of the network. To address these problems, we introduce a point-based changed regional growth (PCRG) module and consistency alignment (CA) constraint into CARGNet, which breaks the limitation of point labels in losing important target details. Specifically, our CARGNet contains two branches: a base decoder branch and an expanded decoder branch. First, we utilize the PCRG module to generate the expanded annotations from the point annotations. Then, the base decoder is supervised by the original point annotations, while the expanded decoder is supervised by the expanded annotations. Finally, the CA constraint is thereby achieved by minimizing the discrepancy between the predictions from both the base and the expanded decoders, which greatly improves the performance of the model. Experimental results on LEVIR-CD-Point and DSIFN-CD-Point datasets demonstrate that our proposed CARGNet can achieve highly competitive results compared with state-of-the-art fully-supervised methods. Code and datasets are available at https://github.com/Wanderlust717/CARGNet.**

***Index Terms*— Change detection, consistency alignment (CA), point annotations, remote sensing, weakly supervised learning.**

I. Introduction

CHANGE detection has long been a focal point in the field of remote sensing image processing. It aims to detect changes in different remote sensing images observing the same geographical areas captured at different times. With the rapid development of Earth observation technologies, change detection has found widespread applications in fields, such as urban development [1], [2], agricultural surveys [3], [4], land cover monitoring [5], [6], [7], disaster assessment [8], and military [9].

Before the prevalence of deep learning, most traditional change detection methods mainly focus on detecting changed pixels and classifying them to generate change maps. A variety of typical machine learning models have been applied to change detection tasks, including support vector machine (SVM) [10], [11], random forests [12], decision trees [13], level set [14], [15], Markov random fields (MRFs) [19], and conditional random fields (CRFs) [16], [17], [18]. Though some specific images can achieve considerable performance with the above methods, they still suffer from low accuracy and lack of generalization ability when faced with complex scenarios and large-scale remote sensing data. In addition, their performance is highly dependent on the extraction of handcrafted features and threshold settings.

With the rapid advancement of deep learning in recent years [20], [21], [22], [23], [24], numerous researchers have adopted the fully supervised deep convolutional neural networks (DCNNs) for change detection, utilizing dense pixel-level annotations to achieve remarkable results. Compared to traditional handcrafted features-based methods [25], DCNNs aim to automatically learn high-level features from varieties of remote sensing data. Based on DCNNs, UNet [26] is widely recognized as one of the standard network architectures in the field of change detection with its ability to effectively capture and leverage context information from both local and global features, achieving accurate and robust change detection results. Daudt et al. [27] proposed a UNet-based Siamese convolutional neural network that uses two remote sensing images captured at different times to detect object changes in the same area. Subsequently, the UNet architecture is extended with some modifications in other works [27], [28], [29], [30]. In this regard, Zhang et al. [28] proposed an end-to-end superpixel-enhanced CD network (ESCNet), by combining differentiable superpixel segmentation and a UNet-based Siamese neural network to mine the different information between images. Peng et al. [29] developed a UNet++ with dense skip connections, which aims to learn more effective features with multiscale semantic information. Papadomanolaki et al. [30] proposed BiDateNet to enhance

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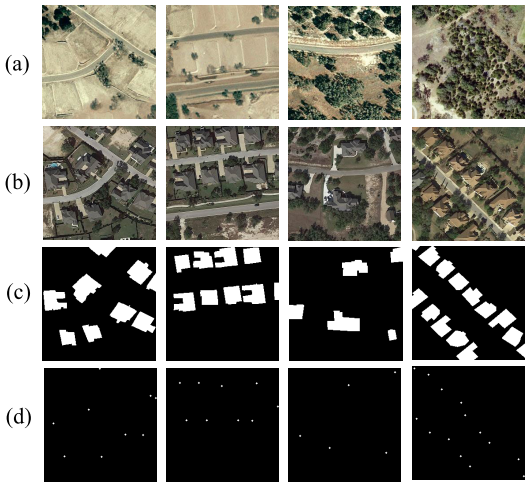


Fig. 1. Comparisons of different levels of annotations. The images from top to bottom are (a) T1 image, (b) T2 image, (c) pixel-level label, and (d) point-level label, respectively.

temporal information between bitemporal images by integrating LSTMs into a UNet structure. Hamdi et al. [31] have developed an algorithm using a modified UNet model for automatic detection and mapping of damaged areas in an ArcGIS environment.

Currently, the existing DCNNs-based change detection methods depend on a large number of annotation samples. However, pairwise labeling large-scale and high-quality bitemporal remote sensing images is very expensive and time-consuming, especially for the large-scale high-resolution remote sensing images [32], [33]. Moreover, unlike semantic segmentation tasks that only need to annotate samples on single-temporal images, the annotation of change detection samples requires simultaneous attention to two or more bitemporal images, which poses significant limitations on the practical applications of change detection techniques in the real world. Therefore, to solve the problem of limited access to annotated samples, researchers begin to explore weakly-supervised learning methods [34], [35], [36], [37], [38] for change detection, which mainly relies on low-level image features (e.g., superpixels or edges) to generate pseudo labels. Among these methods, image-level labels have become the most widely used ones due to their high efficiency in the acquisition. However, change detection in remote sensing images is associated with many challenges [39], such as pseudo-changes, scale changes, and category imbalance, which make it very difficult to retrieve the location of changed objects using only image-level labels. On the contrary, point-level labels can give the approximate location of the changed area and provide more favorable supervision information for the model, as shown in Fig. 1. Nevertheless, point-level labels cannot reveal the complete structure of the changed objects due to their sparsity [40], thus leading to the incomplete prediction within the changed objects.

Based on the above considerations, in this article, we propose a novel point-based weakly supervised learning framework, called as consistency-aligned regional growth network (CARGNet), for the change detection of remote sensing images, which contains two branches: a base decoder branch and an expanded decoder branch. Specifically, we first calculate the mean features of labeled pixels in the changed feature space as the prototypes of the labeled pixels, which allows us to model the similarity of unlabeled and labeled pixels by the distance between the unlabeled pixels and the prototypes in the feature space. In this way, we can expand the annotated regions from the original sparse points to supervise the unlabeled pixels from the perspective of probability. Then, the base decoder is supervised by the original sparse point annotations, while the expanded decoder aims to learn from the expanded annotations. To make a balance between the supervision from the original sparse annotations and the expanded annotations, we further propose a consistency alignment (CA) constraint, aiming to minimize the disparity between the predictions from both the base and the expanded decoders. The proposed CA constraint can support a bidirectional knowledge transfer for both decoders and can effectively control the quality of the point-based changed regional growth (PCRG) module. Note that, point supervision in semantic segmentation typically involves labeling one or more points for an object. In contrast, our proposed point supervision adopts a random dot manner, i.e., randomly assign a point to each changed object in the bitemporal images.

In general, the main contributions of this article are summarized as follows.

1) For the first time, this article proposes a novel task called point-supervised change detection, which is highly valuable in practical applications.
2) We propose a new regional growth change detection framework named CARGNet for point-supervised change detection, which achieves superior performance in remote sensing change detection.
3) We further introduce a CA constraint by minimizing the discrepancy between the predictions from both the base and the expanded decoders to alleviate the issue that the point annotations lack the important details.

The remainder of this article is organized as follows. In Section II, we review related works. In Section III, our proposed method is described in detail. In Section IV, we conduct extensive experiments on the LEVIR-CD-Point and DSIFN-CD-Point datasets and analyze their experimental results. In Section V, we summarize the conclusion of this article.

II. Related Work

A. Deep Learning-Based Change Detection

Change detection, as one of the fundamental tasks in the interpretation of remote sensing images, aims to detect and identify changes in the same area captured at different time points. Recently, a storm involving the field of computer vision has been set off by the emergence of an attention mechanism. Without exception, the change detection task also cannot escape the fate of being infiltrated by the attention mechanism. Peng et al. [41] proposed the difference-enhancement dense-attention convolutional neural network (DDCNN), which captured spatial and temporal dependencies within image pairs by introducing spatial and channel attention. Zhang et al. [42] proposed an attention-based network with a late fusion strategy and used the attention module to adaptively emphasize important features while suppressing

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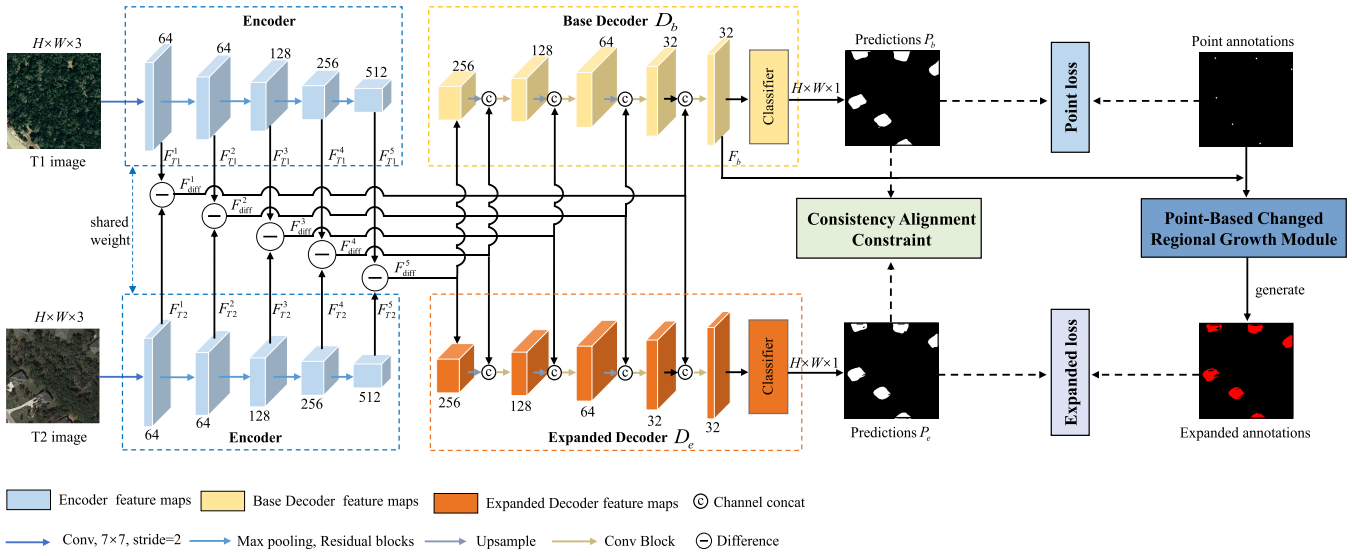


Fig. 2. Overall framework of our proposed CARGNet. CARGNet contains two important components: PCRG module and CA constraint. First, the PCRG module generates the expanded annotations for point annotations. Then, we utilize the predictions P_b, generated by the base decoder D_b, in conjunction with the point annotations, to serve as the point loss for supervising D_b. Similarly, we employ the predictions P_e, produced by the expanded decoder D_e, along with the expanded annotations, to constitute the expanded loss for supervising D_e. Finally, we further let D_b and D_e learn from each other with the CA constraint.

irrelevant features. Zhang et al. [43] employed position- and channel-correlated attentions to extract more effective features for change detection tasks.

With the emergence of the transformer, researchers attempt to incorporate it into the change detection task. Song et al. [44] applied multiple Swin transformer blocks to aggregate the multilevel feature maps of each image captured by the CNN, and then calculate the difference features between the bitemporal images. Similarly, Zhang et al. [45] proposed a pure Swin transformer with a Siamese U-shaped structure (SwinSUNet) to deal with change detection problems, leveraging the Swin transformer block for the encoder, fusion, and decoder components. Chen et al. [46] proposed a transformer-based network with a late fusion strategy to effectively model the spatial–temporal contexts presented in bitemporal images, facilitating the identification of changes of interest while excluding irrelevant alterations. Bandara and Patel [47] utilized a hierarchical transformer encoder in a Siamese architecture with a simple MLP decoder to render multiscale long-range details for change detection.

Although the aforementioned methods have made great progress in the change detection of remote sensing images, training these deep change detectors generally requires a huge number of accurate pixel-level annotations. However, in real remote sensing applications, collecting high-quality labeled data samples tends to require cumbersome manual efforts. Thus, it is of great significance to develop change detection algorithms that can yield satisfactory performance with weak supervision.

B. Weakly Supervised Change Detection

Weakly supervised change detection has also achieved great success, with several methods [35], [36], [37], [38] proposed recently. Most of these methods focus on the most challenging problem by only using image-level labels to detect pixel-level changes [48]. Lu et al. [35] proposed an image-level supervised change detection method based on edge mapping and stacked denoising auto-encoders (SDAEs) network to extract the features of difference from bitemporal images. To highlight relevant changes, Andermatt and Timofte [36] proposed WCDNet that can be trained with image-level semantic labels, which consists of a custom remapping block that enforces a strong separation between background and foreground pixels, a CRF-RNN layer that refines the change mask, and a classifier that predicts the image-level classification for the image pair. Huang et al. [37] introduced a background-mixed augmentation strategy based on image-level labels, aiming to address the issue of limited generalization capability of detectors under diverse environmental changes. Wu et al. [38] proposed a generative adversarial network to detect changed pixels based on the assumption that the changed image pairs can be distinguished into unchanged image pairs by masking out the changed regions. However, it is worth noting that the aforementioned methods may exhibit lower accuracy [34] compared to fully supervised change detection methods, and may present certain challenges in real-world applications.

III. Proposed Method

A. Overall Framework

The overall framework of our proposed CARGNet is shown in Fig. 2, which is composed of two important components: PCRG module and CA constraint. First, the PCRG module generates the expanded annotations for point annotations to provide stronger supervision for the model. Then, the CA constraint is implemented to achieve bidirectional knowledge transfer between the point annotations and the expanded annotations.

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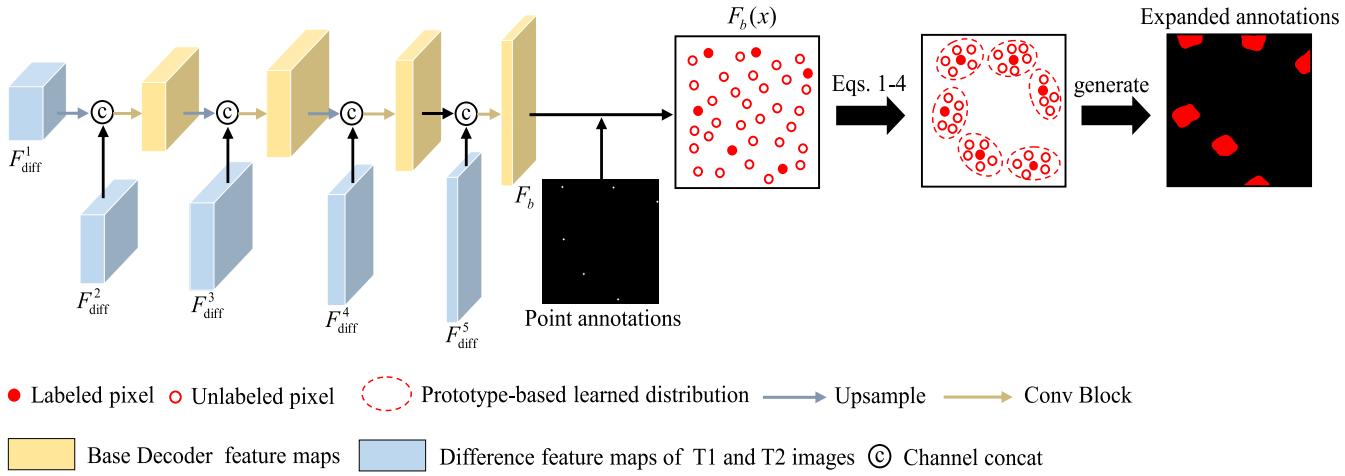


Fig. 3. Illustration of PCRG module.

B. PCRG Module

One of the challenges in weakly supervised change detection is the insufficiency of annotated samples. Therefore, directly using weakly supervised annotations to train the model may lead to poor results. Considering pixels that are close to each other in the feature space are more likely to belong to the same class, a natural idea is to expand the annotated area from the original point annotations.

As shown in Fig. 3, given the difference of multilevel features of T1 image and T2 image $F^i_{\text{diff}} = \{F^1_{\text{diff}}, \ldots, F^5_{\text{diff}}\}$, we can obtain F_b by multilevel differential feature aggregation. After location information is provided by point annotations, we can get the corresponding feature map $F_b(x)$, where $F_b(x)$ represents the changed features of the input pixels x, which are produced from the features before the classification layer of the base decoder. For input pixels x, we can separate them into two parts: labeled pixels x_l and unlabeled pixels x_u. As for the labeled pixels x_l, we build their prototypes by calculating the mean features of labeled pixels x_l, which can be formulated as

$$p_l = \frac{1}{|x_l|} \sum_{\forall x \in x_l} F_b(x) \tag{1}$$

where p_l indicates the prototypes of labeled pixels x_l. After obtaining p_l, we can calculate the distance d between labeled and unlabeled pixels by the following formula:

$$d = F_b(x_u) - p_l. \tag{2}$$

Recall that our goal is to expand the annotated regions. To this end, we define an expanded label matrix E, and first initialize E with the point annotations y_l

$$E_{(l)} \leftarrow y_l \tag{3}$$

where $E_{(l)}$ represents the value in the expanded label matrix at labeled pixels x_l and "$\leftarrow$" denotes the right-to-left assigning operator. For unlabeled pixels in E, we update its label $E_{(u)}$ with the following formula:

$$E_{(u)} = e^{-d}. \tag{4}$$

In this way, we can ensure that the values of the expanded label matrix E range from 0 to 1. These expanded annotations are in the form of soft scores, denoting the probability that each pixel x belongs to the changed class.

It is worth noting that the generation process of our expanded annotations is dynamically adjusted since the parameters of the base decoder are progressively optimized during training, enabling us to obtain more accurate prototypes of labeled pixels.

C. CA Constraint

When we obtain the expanded annotations, a natural idea is to replace the original point labels with the expanded annotations to train the model. With more labeled samples in the expanded annotations, the model can receive stronger supervision, which in turn helps to produce more accurate expansions in growing regions. However, due to the potential errors contained in the expanded annotations, learning directly from them may mislead the training of the model, leading to worse detection results. In this article, instead of training with the expanded annotations directly, we propose a CA constraint in which an expanded decoder is used to extract the supervised information contained in expanded annotations.

Formally, we let D_b denote the base decoder and D_e denote the expanded decoder. Both decoders share the same encoder network. Then, we utilize original point annotations to supervise the predictions P_b generated by the base decoder D_b. We first define the point loss $\mathcal{L}_p$ with the binary cross entropy for D_b as

$$\mathcal{L}_p = -\frac{1}{n_p} \sum_{i=1}^{n_p} y_l^{(i)} \log\left(P_b^{(i)}\right) \tag{5}$$

where n_p denotes the number of annotated pixels in the original point annotations y_l. $y_l^{(i)}$ and $P_b^{(i)}$ represent the probability of the changed class at pixel i predicted by the base decoder D_b, and the label for the pixel i, respectively.

Similarly, the predictions P_e generated by the expanded decoder D_e can be supervised by the expanded annotations E. Considering the severe class imbalance problem in change

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detection, we define the expanded loss $\mathcal{L}_e$ by the binary cross entropy loss with weights for D_e as

$$\begin{cases} \mathcal{L}_e = -\dfrac{1}{n_e}\displaystyle\sum_{i=1}^{n_e}\sum_{j=1}^{C} w_j E^{(i,j)} \log\left(P_e^{(i,j)}\right) \\ w_1 = \dfrac{n_1+n_2}{2n_1} \\ w_2 = \dfrac{n_1+n_2}{2n_2} \end{cases} \tag{6}$$

where n_e denotes the number of annotated pixels in the expanded annotations E, C represents the number of categories, and w_j indicates the weight of the jth class. w_1 and w_2 correspond to the weights of negative samples and positive samples, respectively. n_1 and n_2 denote the number of negative and positive samples, respectively. With the expanded loss in (6), the expanded decoder D_e can gradually get supervised information from the expanded annotations.

However, there may exist some potential errors inevitably in the expanded annotations, and learning directly from them may mislead the training of the network. To this end, we further propose a CA constraint $\mathcal{L}_c$ by minimizing the discrepancy between the predictions from both the base and the expanded decoders

$$\mathcal{L}_c = -\frac{1}{n}\sum_{i=1}^{n}\left\| P_b^{(i)} - P_e^{(i)} \right\|^2 \tag{7}$$

where n indicates the number of pixels in the whole image. Through CA constraint $\mathcal{L}_c$, we can let the point annotations P_b and the expanded annotations P_e achieve online self-supervision.

The constraint $\mathcal{L}_c$ offers twofold benefits. First, although the expanded annotations can alleviate the lack of labeled samples issue, they may inherently contain errors and noise. By contrast, the original point-level annotations are accurate but sparse. Therefore, the constraint in (7) achieves a balance between both types of annotations, combining their strengths. In addition, $\mathcal{L}_c$ can be viewed as a form of soft knowledge distillation. By leveraging the expanded decoder D_e, the base decoder D_b does not directly learn from the expanded annotations. Instead, it is guided by the distilled knowledge obtained from D_e, which is a more moderate way to utilize the valuable information presented in the expanded annotations.

Therefore, the overall loss function $\mathcal{L}$ in our CARGNet framework is formulated as follows:

$$\mathcal{L} = \mathcal{L}_p + \mathcal{L}_e + \mathcal{L}_c. \tag{8}$$

With the collaborative optimization of $\mathcal{L}_p$, $\mathcal{L}_e$, and $\mathcal{L}_c$, we can learn more discriminative feature differences from bitemporal remote sensing image pairs.

It is worth noting that the expanded decoder is employed only during training, which is discarded in the test phase. In other words, we just take the output of the base decoder as our final prediction results. The reason for discarding the expanded decoder is as follows. Discarding the expanded decoder during testing can reduce model complexity and computational cost, thereby improving model testing efficiency, as shown in Table I.

TABLE I
INFLUENCE OF THE EXPANDED DECODER ON MODEL EFFICIENCY DURING THE TESTING PHASE

	Params. (M)	Test time (s)
with expanded decoder	19.10	66
w/o expanded decoder	16.73	56

TABLE II
NUMBERS OF THE LABELED PIXELS IN POINT-LEVEL AND PIXEL-LEVEL ANNOTATIONS USED IN LEVIR-CD-POINT AND DSIFN-CD-POINT DATASETS

Dataset name	Point-level annotations	Pixel-level annotations
LEVIR-CD-Point	339,330	21,412,971
DSIFN-CD-Point	380,681	330,013,136

IV. EXPERIMENTS

A. Datasets Description

1) LEVIR-CD-Point: The creation of the LEVIR-CD-Point dataset relies on the LEVIR-CD [49] dataset. The LEVIR-CD-Point dataset is a large-scale remote sensing change detection dataset, covering 20 different regions that sit in several cities in Texas, USA. It consists of 637 high spatial resolution bitemporal remote sensing image pairs with a size of 1024 × 1024 pixels, which were collected from the Google Earth platform. For annotation, this dataset provides a total of 31 333 change labels of building instances. This dataset includes not only building updating but also building decline for more general building changes. There are 445/64/128 pairs for training/validation/testing. Following the previous works, we crop the original images into image patches of 256 × 256 pixels without overlapping and then obtain 7120/1024/2048 pairs of patches for training/validation/testing, respectively. The point annotations in our LEVIR-CD-Point dataset contain 339 330 pixels for model training. The LEVIR-CD-Point dataset will be made publicly available at https://github.com/Wanderlust717/CARGNet.

2) DSIFN-CD-Point: The creation of the DSIFN-CD-Point dataset is based on the DSIFN-CD [42] dataset. The DSIFN-CD-Point dataset is a general change detection dataset of Google Earth images, which contains six large bitemporal high-resolution images that are captured from six major cities in China. This dataset contains the changes in different land cover objects, such as roads, buildings, farmland, and water bodies. It contains different brightness and contrast images and different land objects from six cities, which makes it difficult to detect changed areas. There are 3600/340/48 pairs for training/validation/testing. Following the previous works, we crop the 512 × 512 images into image patches of 256 × 256 pixels without overlapping and then obtain 14 400/1360/192 pairs of patches for training/validation/testing, respectively. The point annotations in our DSIFN-CD-Point dataset contain 380 681 pixels for model training. The DSIFN-CD-Point dataset will be made publicly available at https://github.com/Wanderlust717/CARGNet.

Table II lists the number of labeled pixels with point-level annotations and pixel-level annotations in both datasets. From

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the table, we can find that point-level annotations exhibit a magnitude of several orders less than pixel-level annotations.

B. Evaluation Metrics

To quantitatively analyze the performance of the models, we take advantage of the $F1$-score (F_1), the overall accuracy (OA), and the intersection over union (IOU) to evaluate the performance of the models. Each of these metrics provides a different perspective on the performance evaluation of a model.

We use the $F1$-score of the change category as the main evaluation metric, which is calculated as follows:

$$F_1 = \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}} \tag{9}$$

where precision denotes the proportion of all predicted positive samples that actually belong to the number of positive samples and recall denotes the proportion of positive samples that are predicted to be correct in the number of all positive samples. They can be formulated as

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{10}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{11}$$

IOU is a common metric used to evaluate the degree of overlap between the predicted segmentation map and the ground truth. In the change detection task, IOU for the change category can be calculated as follows:

$$\text{IOU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{12}$$

OA is a measure of the proportion of correctly classified samples out of the total samples. It is calculated as

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}} \tag{13}$$

where TP, FN, FP, and TN represent the number of true positive samples, false negative samples, false positive samples, and true negative samples, respectively.

C. Experimental Details

We implement our model based on the PyTorch framework [50], using a single NVIDIA GeForce RTX 3090Ti GPU with 24G memory for training and testing. We employ the ResNet-18 model [51] pretrained on ImageNet [52] as the feature encoding network. During the training phase, we apply data augmentation through autocontrast, equalize, posterize, solarize, brightness, and sharpness. We use the stochastic gradient descent (SGD) optimizer with an initial learning rate of 0.001, a momentum of 0.9, and a weight decay of 5e-4 to optimize the model parameters. We adopt the fixed-step learning rate decay strategy, where the initial learning rate is multiplied by 0.9 at every 1000 iterations. We train models for 40k iterations for LEVIR-CD-Point and the batch size is set to 8. The experimental settings of the DSIFN-CD-Point dataset are consistent with the LEVIR-CD-Point dataset. Validation is performed after every 1000 iterations, and the best model on the validation set is used for evaluation on the test set.

D. Comparative Method

In this section, to verify the effectiveness of our proposed CARGNet, we make a comprehensive comparison between our proposed approach and several state-of-the-art methods in this field. The compared methods involve weakly supervised and fully supervised change detection methods, including FCD-GAN [38], FCD-GAN [38] with BGMix [37], three convolution-based methods (FC-EF [27], FC-Siam-Diff [27], and FC-Siam-Conc [27]), DTCDSCN [2], transformer-based method (BIT [46]), and USSFCNet [53].

1) FCD-GAN [38]: A fully convolutional change detection framework, which incorporates unsupervised, weakly supervised, region supervised, and fully supervised change detection into one framework.

2) BGMix [37]: A background-mixed augmentation strategy, cooperating with augmented and real data consistency loss to improve the generalization performance of the model under complex background changes.

3) FC-EF [27]: An image-level fusion network, where the bitemporal images are concatenated as a single input to a ConvNet to detect changes.

4) FC-Siam-Diff [27]: A feature-difference network, which employs a Siamese FCN to extract multilevel features and utilizes feature differences to detect changes.

5) FC-Siam-Conc [27]: A feature-concatenation network, which leverages a Siamese FCN to extract multilevel features and uses feature concatenation as a fusion strategy for bitemporal information.

6) DTCDSCN [2]: An attention-based method, which utilizes a dual attention module (DAM) to exploit the interdependencies between channels and spatial positions to improve the feature representation.

7) BIT [46]: A transformer-based method, which utilizes a transformer to model the context present in bitemporal images more effectively.

8) USSFCNet [53]: An ultralightweight spatial–spectral feature cooperation network, which designs a multiscale decoupled convolution to capture multiscale features of changed objects and introduces a spatial–spectral feature cooperation strategy to refine features by comodeling spatial and spectral features.

E. Experimental Results Analysis

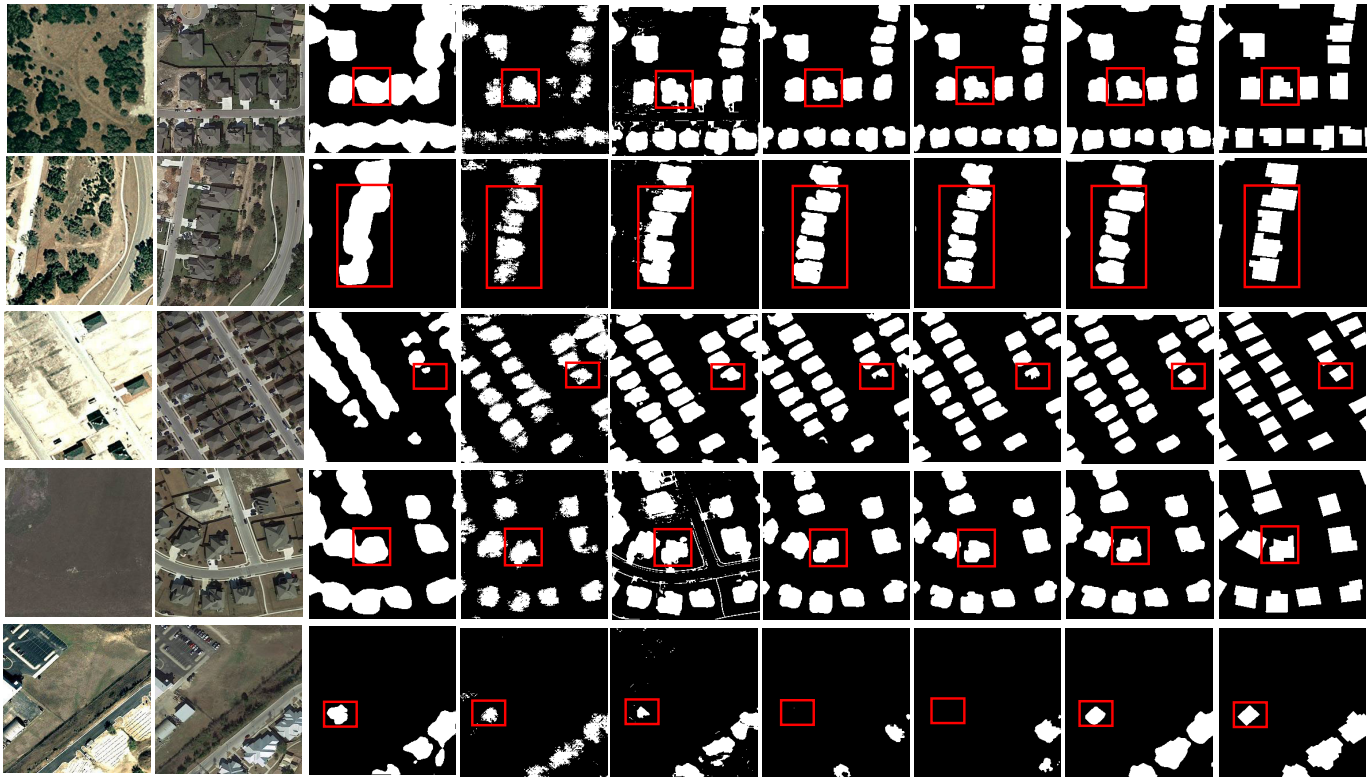
Table III shows the results of different change detection methods on the LEVIR-CD-Point and DSIFN-CD-Point test sets.

1) Evaluation on the LEVIR-CD-Point Dataset: For the LEVIR-CD-Point dataset, from the table, it can be seen that our proposed CARGNet can yield a good result with 83.81% in $F1$, 72.13% in IoU and 98.28% in OA, which is very close to those obtained from the state-of-the-art fully supervised method. Moreover, our proposed method can even outperform the classical fully supervised methods FC-EF and FC-Siam-Conc by 0.41% and 0.12% points in $F1$ score. To further verify the effectiveness of our proposed CARGNet, it is necessary to compare it with other weakly supervised change detection methods. Here, we first conduct image-level

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TABLE III
QUANTITATIVE COMPARISON RESULTS OF DIFFERENT CHANGE DETECTION METHODS ON LEVIR-CD-POINT AND DSIFN-CD-POINT TEST SETS

Method	Supervision	LEVIR-CD-Point					DSIFN-CD-Point				
		Precision	Recall	F1	IoU	OA	Precision	Recall	F1	IoU	OA
FC-EF [27]	pixel-level	86.91	80.17	83.40	71.53	98.39	**72.61**	52.73	61.09	43.98	88.59
FC-Siam-Diff [27]	pixel-level	89.53	83.31	86.31	75.92	98.67	59.67	65.71	62.54	45.50	86.63
FC-Siam-Conc [27]	pixel-level	**91.99**	76.77	83.69	71.96	98.49	66.45	54.21	59.71	42.56	87.57
DTCDSCN [2]	pixel-level	88.53	86.83	87.67	78.05	98.77	53.87	77.99	63.72	46.76	84.91
BIT [46]	pixel-level	89.24	89.37	89.31	80.68	98.92	68.36	70.18	69.26	52.97	89.41
USSFCNet [53]	pixel-level	88.10	**92.01**	**90.01**	**81.84**	**99.00**	62.18	**78.99**	**69.59**	**53.36**	**90.76**
FCD-GAN [38]	image-level	**70.86**	47.41	56.81	39.67	96.33	**64.81**	28.70	39.78	24.83	**85.24**
FCD-GAN + BGMix [37]	image-level	67.46	**69.08**	**68.26**	**51.81**	**96.73**	54.67	**44.37**	**48.99**	**32.44**	84.30
DTCDSCN [2]	point-level	49.65	43.92	46.61	30.38	94.20	52.13	32.71	40.20	25.16	73.65
USSFCNet [53]	point-level	56.10	74.85	64.13	47.20	96.80	77.00	33.11	46.31	30.13	69.66
BIT [46]	point-level	**85.86**	62.08	72.06	56.32	96.61	**87.07**	32.16	46.97	30.69	66.59
FCD-GAN [38]	point-level	71.76	84.61	77.66	63.48	97.52	55.94	52.20	54.00	36.99	84.89
CARGNet (ours)	point-level	80.57	**87.32**	**83.81**	**72.13**	**98.28**	57.57	**72.94**	**64.35**	**47.44**	**86.27**



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 4. Qualitative results of different change detection methods on LEVIR-CD-Point dataset: (a) T1 image, (b) T2 image, (c) BIT (point-level supervision), (d) USSFCNet (point-level supervision), (e) FCD-GAN (point-level supervision), (f) FC-EF (pixel-level supervision), (g) FC-Siam-Conc (pixel-level supervision), (h) CARGNet (Ours), and (i) ground truth.

supervised change detection experiments on the LEVIR-CD-Point dataset. Compared with FCD-GAN with BGMix, our proposed CARGNet can bring about 20.32% IoU improvements. Then, we further conduct point-supervised change detection experiments on the LEVIR-CD-Point dataset. Since there is no point-based change detection method yet, for a fair comparison, we define a baseline method for comparisons: we use point annotations to train a standard change detector. Experimental results show our CARGNet outperforms the baseline methods by a large margin. Specifically, CARGNet outperforms FCD-GAN and BIT by 8.65% and 15.81% points in IoU, respectively.

The visualization results of the comparative methods are shown in Fig. 4. As highlighted in the red rectangle, due to the bitemporal spectral changes caused by changes in land cover and seasonal illumination, there will be some areas of pseudo-changes and missed detections in all compared methods. As shown in Fig. 4(c)–(g), there are notable omissions and false alarms in the change maps generated by the compared methods. By contrast, our proposed method effectively mitigates the influence of noise and pseudo-changes while preserving the internal compactness of changed objects. In addition, the results of the compared methods show that the edges of building objects are blurred. In contrast, CARGNet

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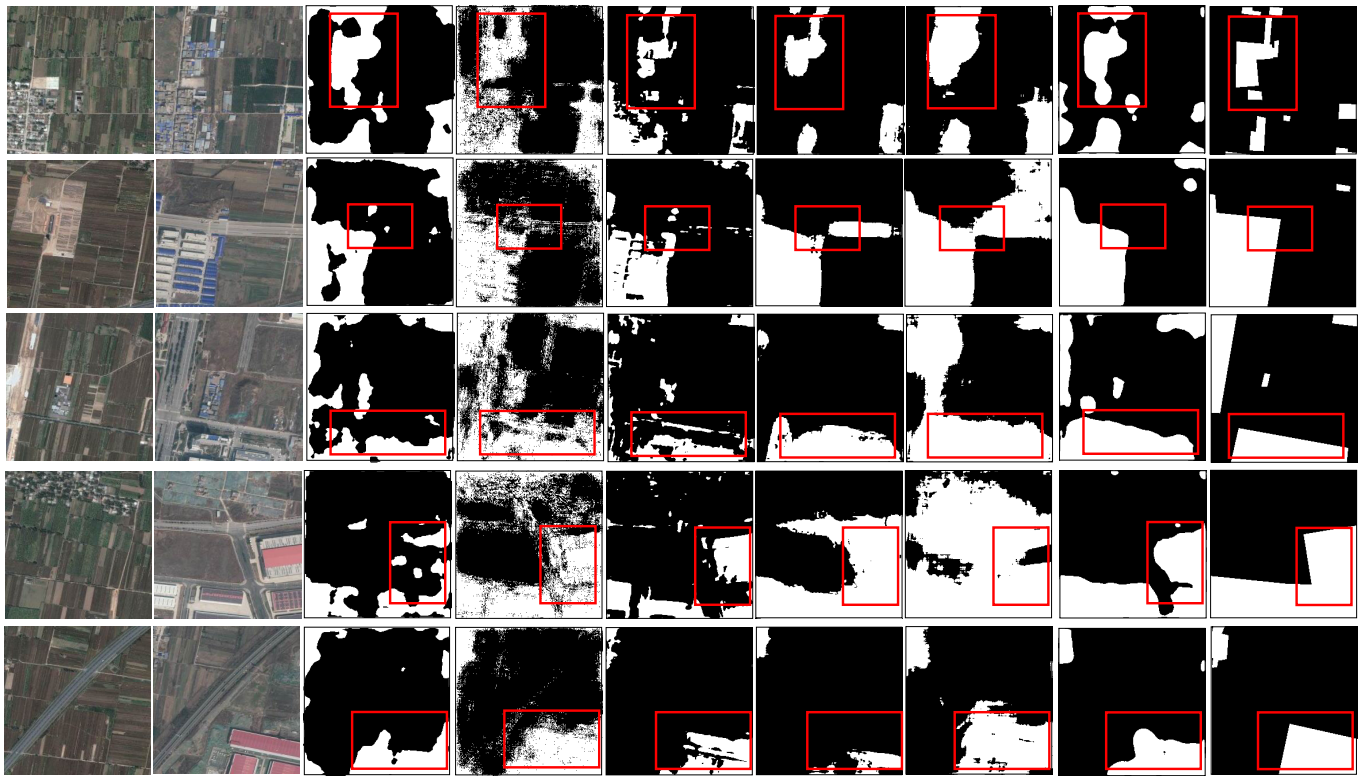


Fig. 5. Qualitative results of different change detection methods on DSIFN-CD-Point dataset: (a) T1 image, (b) T2 image, (c) BIT (point-level supervision), (d) USSFCNet (point-level supervision), (e) FCD-GAN (point-level supervision), (f) FC-EF (pixel-level supervision), (g) FC-Siam-Conc (pixel-level supervision), (h) CARGNet (Ours), and (i) ground truth.

can detect changed objects and preserve the details of their boundaries more accurately.

2) Evaluation on the DSIFN-CD-Point Dataset: Compared with the previous LEVIR-CD-Point dataset, DSIFN-CD-Point contains the multiscale and irregular changes in six major cities' landcover objects, which makes it quite challenging for change detection algorithms to detect precise changes. It can be seen from Table III that the performance of most methods on DSIFN-CD-Point is significantly worse than that of LEVIR-CD-Point. From Table III, we can also observe that our proposed method surpasses many state-of-the-art fully supervised methods (e.g., FC-EF, FC-Siam-Diff, FC-Siam-Conc, and DTCDSCN) in $F1$ and IoU. Compared with point-supervised methods, our CARGNet outperforms FCD-GAN and BIT by 10.35% and 17.38% points in $F1$, respectively. These results demonstrate the effectiveness of the proposed method in dealing with complex DSIFN-CD-Point datasets.

The qualitative results of the different change detection methods on the DSIFN-CD-Point dataset are depicted in Fig. 5. As can be seen, the DSIFN-CD-Point dataset contains intricate urban structures and diverse land cover types, which may lead to salt-and-pepper noise and misclassifications in the change detection maps of the compared methods, especially at the boundaries of irregular objects, as shown in Fig. 5(c)–(g). By contrast, CARGNet better retains the intricate features of objects along with their boundaries, exhibiting enhanced resistance to noise. Consequently, the change maps produced by CARGNet are closer to the ground truth, surpassing the outcomes of other approaches, as illustrated in Fig. 5(h).

F. Comparison of Model Efficiency

To ensure a fair comparison of model efficiency, we test all the methods on a computing server equipped with an Intel Xeon E5-2678 v3 CPU and an NVIDIA GeForce RTX 3090Ti GPU. Table IV lists the number of model parameters (Params.) and GPU test time of different methods evaluated on the LEVIR-CD-Point test set, using input images sized at $256 \times 256 \times 3$. Experimental results show that our proposed CARGNet outperforms DTCDSCN, FCD-GAN, and FCD-GAN + BGMix with smaller model parameters and less computational cost.

G. Ablation Study

To verify the effectiveness of each module in the proposed CARGNet, we conduct ablation studies on the LEVIR-CD-Point and DSIFN-CD-Point datasets with different settings, as shown in Tables V and VI. From the two tables, it can be found that our proposed method can achieve a remarkable improvement compared with the baseline method using $\mathcal{L}_p$ only.

For the LEVIR-CD-Point dataset, we can observe that when adding the loss $\mathcal{L}_e$ on the basis of the loss $\mathcal{L}_p$, the method can achieve a performance of 68.63% in $F1$ and 52.24% in IoU, which corresponds to a significant improvement of 10.75%

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TABLE IV
COMPARISON OF MODEL EFFICIENCY. WE REPORT THE NUMBER OF MODEL PARAMETERS (PARAMS.) AND TEST TIME ON THE LEVIR-CD-POINT TEST SET

Method	Backbone	Supervision	Params. (M)	Test time (s)
FC-EF [27]	UNet	full	1.35	24
FC-Siam-Diff [27]	UNet	full	1.35	25
FC-Siam-Conc [27]	UNet	full	1.55	31
DTCDSCN [2]	SE-Res34	full	41.07	60
BIT [46]	ResNet18	full	3.55	58
USSFCNet [53]	UNet	full	1.52	146
FCD-GAN [38]	UNet	weak	43.35	61
FCD-GAN + BGMix [37]	UNet	weak	43.35	61
CARGNet (ours)	ResNet18	weak	19.10	56

TABLE V
ABLATION STUDY PERFORMED ON THE LEVIR-CD-POINT DATASET

$\mathcal{L}_p$	$\mathcal{L}_e$	$\mathcal{L}_c$	F1	IoU	OA
✓			57.88	40.73	96.68
✓	✓		68.63	52.24	96.88
✓	✓	✓	83.81	72.13	98.28

TABLE VI
ABLATION STUDY PERFORMED ON THE DSIFN-CD-POINT DATASET

$\mathcal{L}_p$	$\mathcal{L}_e$	$\mathcal{L}_c$	F1	IoU	OA
✓			43.74	27.99	58.20
✓	✓		49.80	33.15	67.56
✓	✓	✓	64.35	47.44	86.27

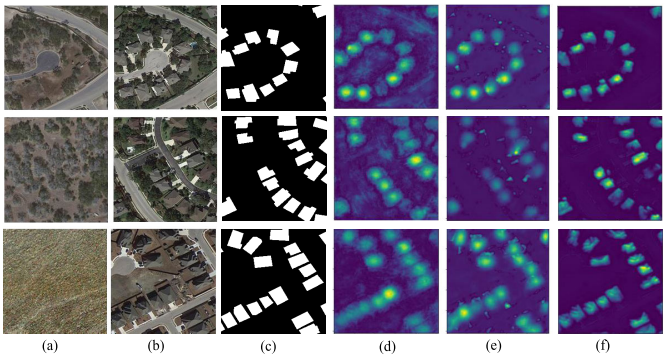


Fig. 6. Feature visualization results of different experimental settings. (a) T1 image. (b) T2 image. (c) Ground truth. (d) Baseline. (e) Baseline + PCRG. (f) CARGNet.

points in $F1$ and 11.51% points in IoU when comparing to the baseline. Moreover, as we incorporate consistency loss $\mathcal{L}_c$, the performance of our network is further enhanced, yielding remarkable increases of 25.93% points in $F1$ and 31.4% points in IoU when compared to the baseline.

In addition to the baseline loss $\mathcal{L}_p$, by introducing the loss $\mathcal{L}_e$, we observe notable performance enhancements in the DSIFN-CD-Point dataset. Specifically, our method achieves a remarkable improvement, with $F1$ score of 49.80% and OA of 67.56%, which represents a significant increase of 6.06% points in $F1$ and 9.36% points in OA, compared to the baseline. Furthermore, by incorporating the consistency loss $\mathcal{L}_c$, the performance of our network is further enhanced, with extraordinary improvements of 20.61% points in $F1$ and 19.45% points in IoU compared with the baseline method. These results highlight the effectiveness of our approach in improving change detector performance.

To visually analyze the impact of each module in the proposed method, we visualize the heatmap of intermediate features. Note that, we take the features before the classification layer of the base decoder for visualization. Fig. 6 shows the feature visualization of different experimental settings. From the visualization results of the feature map, it can be found that our method can identify the boundaries of objects more clearly and maintain the separability between objects.

V. CONCLUSION

In this article, we introduced point labels into remote sensing change detection for the first time. Based on this, we proposed a simple yet effective framework, called as CARGNet, for weakly supervised change detection in remote sensing images. Our proposed CARGNet exploited a prototype distance metric between unlabeled and labeled pixels in the feature space to expand point annotations, effectively addressing the challenge of the limited availability of labeled samples. In addition, the CARGNet leveraged a consistent alignment constraint to facilitate bidirectional knowledge transfer between point annotations and expanded annotations, which mitigated the issue of point annotations losing intricate details within the target context. Experiments performed on the LEVIR-CD-Point dataset and DSIFN-CD-Point dataset demonstrated the proposed CARGNet can achieve remarkably competitive results compared with state-of-the-art fully-supervised change detection methods.

Currently, there is a notable absence of research investigating point-supervised change detection in remote-sensing images. We hope that our proposed method will serve as a strong baseline and help future research in point-supervised change detection. In the future, we will follow up to investigate the proficiency of point-supervised change detection tasks in remote sensing images and further explore more effective weakly supervised change detection methods for point annotations. In addition, we will also attempt to integrate multimodal data into weakly supervised change detection to fully mine data information and improve detection performance.

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